

ASSINGMENT 4

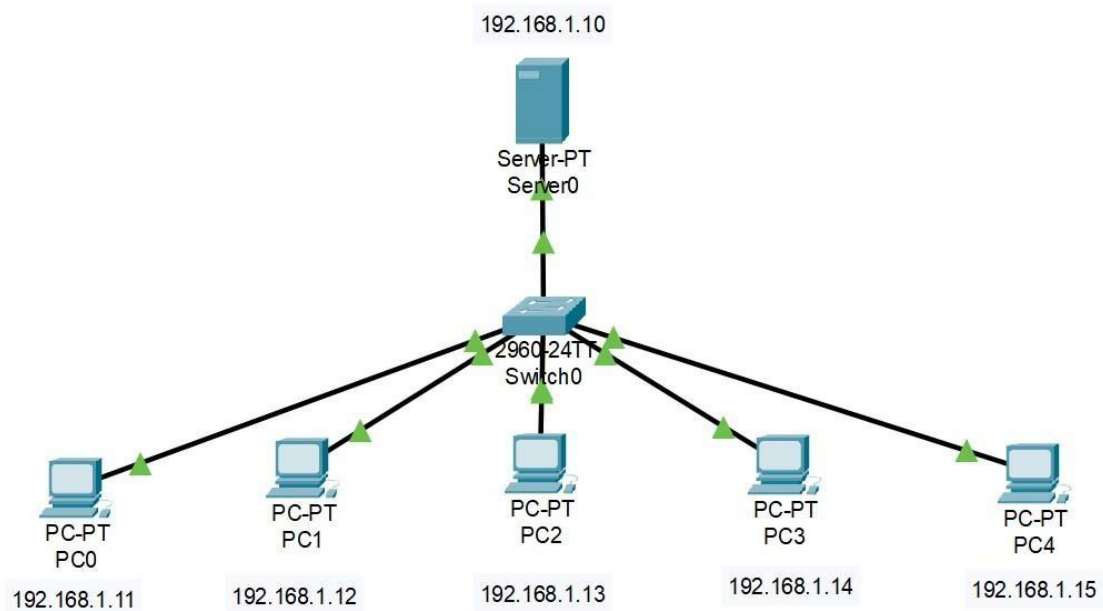
NAME : MOHAMED ANAS A

ROLL NO : 23CS090

CLASS : II CSE B

SUBJECT : COMPUTER NETWORKS

TITLE : THREE WAY HANDSHAKE



Title: Robust Three-Way Handshake Network Design

Objective: Design a reliable network to ensure robust three-way handshake connection establishment under high client load.

Tool Used: Cisco Packet Tracer (CPT)

Step 1: Open Cisco Packet Tracer.

Step 2: Drag and drop the following devices:

- 1 Server (Server-PT)
- 1 Switch (2960-24TT)
- 5 PCs (PC-PT)

Step 3: Connect all devices:

- Use copper straight-through cables.
- Connect each PC and Server to the Switch.

Step 4: Configure IP addresses:

- Server0: 192.168.1.10

- PC0: 192.168.1.11
- PC1: 192.168.1.12
- PC2: 192.168.1.13
- PC3: 192.168.1.14
- PC4: 192.168.1.15
- Subnet mask: 255.255.255.0 for all devices.

Step 5: Simulate three-way handshake:

- Use TCP communication in simulation mode.
- Initiate a connection from each PC to Server0.

Step 6: Verify handshake:

- Capture SYN, SYN-ACK, and ACK packets.
- Ensure successful handshake under multiple simultaneous requests.

Step 7: Test Network Reliability:

- Use the ping command from each PC to Server0.
- Confirm there is no packet loss.